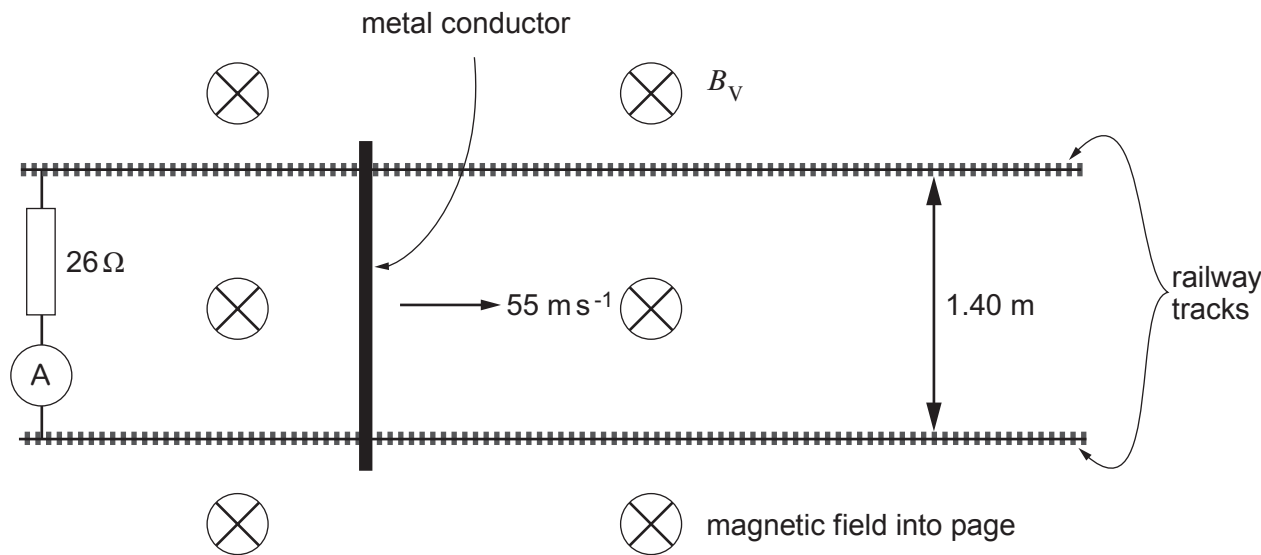


5. A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component (B_v) of the Earth's magnetic field.



- (a) (i) Explain why a current is detected by the ammeter. [2]

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- (ii) State why the current is independent of the horizontal component of the Earth's magnetic field. [1]

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- (b) Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is $184 \mu\text{A}$. [3]

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(c) State the direction of the current in the resistor and how you obtained this direction. [2]

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(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

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